

Institutions for Comparative Discovery in Scaling Markets

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October 2026 · Journal article draft

Abstract

Scaling technologies can concentrate production without determining the efficient organization of comparative evaluation. This article analyzes institutions that assemble evidence about feasible alternatives when adaptive allocation limits the comparisons generated by ordinary market operation. It distinguishes productive capability, entry feasibility, comparative evidence, and appropriation. A joint-permission model shows how separately controlled, informationally complementary inputs can remain unavailable even when a calibrated experiment has positive social value. A committed contract is participation-feasible when gross decision value covers provision and coordination costs; implementation requires coordination as well as commitment. The institutional comparison considers private contracting, public procurement, independent testing facilities, and shared infrastructures while retaining productive scale, privacy, security, and enforcement costs. A prospective AI-service evaluation design connects these arrangements to bounded decisions rather than imagined firms. The comparison-budget program motivates the inquiry but supplies neither a monetary price of evidence nor an empirical verdict on platform taxation. The contribution is an explicit route from an adaptive-learning diagnosis to the governance and financing of a worthwhile comparative experiment.

Keywords: institutions, comparative discovery, adaptive markets, contestability, information value.

JEL codes: D02, D83, L41, H41.

1. Introduction

Technologies with large fixed costs, shared infrastructures, and cumulative learning can support a small number of highly capable producers. That fact does not settle whether the institutions governing evaluation allow feasible alternatives to demonstrate their value. Production and evaluation can have different efficient scales. A concentrated production technology can coexist with independent testing; a fragmented supplier population can depend on one common ranking and feedback system. Institutional analysis should therefore examine the rules that produce comparative evidence alongside the rules that allocate current output.

This article develops a theory of institutions for comparative discovery in scaling markets. Its motivating hypothesis is that some access arrangements suppress socially valuable experiments even when the incumbent's productive advantages are genuine. The hypothesis is conditional. An unrealized firm is not automatically a social loss, and an additional comparison is not automatically useful. A feasible experiment must improve a consequential decision sufficiently to justify its costs. The question is whether the rules controlling the required resources, permissions, and disclosures permit that experiment to occur.

Shadow Futures introduces a comparison budget for a specified adaptive allocation experiment and shows that, under stated assumptions, finite comparison can prevent universal recovery of nonconstant contribution functionals from one growing history (Erlic, 2026). This statistical result motivates examining the institutional production of evidence. It does not by itself establish inefficient exclusion, a welfare loss, or an optimal tax. The institutional bridge requires a decision problem, an informative feasible experiment, and an account of why private incentives fail to supply it.

The article supplies that bridge through a simple joint-permission model. Two separately controlled inputs are complementary in producing decision-relevant evidence. Each holder can prefer withholding even when the combined experiment has positive social value. A committed and enforceable joint contract is participation-feasible when its total value covers the permission and coordination costs. Implementation also requires a contracting sequence that resolves coordination. The result is an application of familiar appropriation and complementarity problems, not a new general theorem about institutions. Its contribution is to connect those problems to a clearly specified evidence-producing activity rather than to treat market plurality as an automatic proxy for discovery.

The institutional implications concern the allocation of evaluation rights, obligations, and procedures. Possible arrangements include private contracting, public procurement trials, independent facilities, and shared infrastructure. Each has different information, enforcement, and cost requirements. The article also gives a bounded AI-service evaluation design and specifies evidence gates linking the statistical program to policy. No actual platform is assessed and no claim is made that contemporary AI firms pay less than an empirically established optimal tax.

2. Institutions, discovery, and the relevant economic object

Institutions matter here as rules and procedures governing who can propose an experiment, qualify for access, allocate test opportunities, record outcomes, evaluate claims, and disclose findings. This use is consistent with the rule-oriented accounts of North (1990) and Hodgson (2006). It identifies a class of phenomena that can be studied empirically: eligibility criteria, contractual permissions, procurement terms, sampling protocols, record-retention rules, and

appeal procedures. A platform company is an organization participating in this system, not a synonym for the institution as a whole.

Hayek's (1945) analysis of dispersed knowledge makes knowledge coordination central to economic organization. It does not establish that every allocation mechanism generates all the information needed to evaluate its own alternatives. The present inquiry addresses the conditions under which such alternatives become observable. Comparative discovery requires more than a stock of dispersed facts: it can require an intervention, a qualified prototype, or access to a resource jointly controlled by actors who do not capture the resulting decision value. Institutions determine whether these opportunities can be assembled.

Contestable-market theory provides another relevant starting point. Baumol (1982) emphasizes the importance of potential competition rather than equating competition with the number of incumbents. In scaling technologies, however, an alternative's potential effectiveness may need to be demonstrated before financing or adoption is available. The relevant institutional question is whether a technically and financially bounded project can obtain an informative evaluation route. This question complements entry conditions; it does not eliminate the need to establish whether entry is economically feasible.

Increasing returns and endogenous sunk costs further discipline the argument. Arthur (1989) analyzes historical lock-in with increasing returns; Sutton (1991) explains why capability expenditure can sustain concentration as markets expand. These contributions warn against assuming that a larger market should support arbitrarily many effective firms. The current account therefore distinguishes efficient production scale from the scale and independence of evaluation. Preserving a comparison route need not mean duplicating an entire frontier production system.

Private appropriation is a separate issue. Arrow (1962) analyzes the allocation of resources to invention under uncertainty, while Jones and Tonetti (2020) examine the economics of nonrival data. The information used in a decision can benefit actors beyond the party that supplies the experiment. Yet a given test may consume scarce compute, staff time, secure access, or attention. Nonrival use of its result does not make its production free. An institutional proposal must identify both the appropriability gap and the real costs of assembling the experiment. Finally, complementary permissions can create a blocking problem. Heller and Eisenberg (1998) analyze an anticommons problem in biomedical research. The analogy here concerns separate control over jointly necessary inputs to comparative evidence. It is not a claim that all platform access restrictions are anticommons or that removing property rights is efficient. The relevant variables are the joint experiment's value, the holders' opportunity costs, the benefits they retain, and the transaction costs of a lawful agreement.

3. Productive scale and the supply of evidence

Four objects organize the analysis. Productive capability concerns the resources and technology needed to supply a defined service. Feasible entry concerns a project's ability to cover costs and obtain the required resources. Comparative evidence concerns observations that discriminate among models relevant to a specified decision. Appropriation concerns which parties receive the experiment's decision benefits and bear its costs. A failure in one object cannot be inferred solely from an observation about another.

For example, a capability threshold may imply that only three producers can serve a frontier task at the available resource scale. That does not show that independent evaluation is infeasible. A small number of producers might submit systems to multiple independent task cohorts.

Conversely, twenty nominal suppliers using a common deployment gate can remain dependent

on one evidence channel. Supplier counts therefore describe organizations without necessarily describing the statistical independence or coverage of the experiments through which their claims are evaluated.

In the foundational allocation experiment, the comparison budget sums conditional probability outside the current leader. Its role is diagnostic within that model: it can bound allocation information under bounded profiles and support a sufficient condition for a learning obstruction. It is neither a concentration index nor a universally fungible quantity of evidence.

$$\varepsilon_t = 1 - \max_i p(i, t), \quad B(T) = \sum_{t < T} \varepsilon_t. \quad (1)$$

A concentrated allocation can have an infinite budget. An infinite budget can also be uninformative about a parameter if the relevant profiles do not vary. Additional observed responses can identify a mechanism that allocation alone cannot identify. The complete-history impossibility result further needs positive finite-history kernels, common predictable profiles and state rules, the specified history information, and a Hellinger domination condition. A small finite-horizon estimated budget is not evidence that an actual market satisfies all these assumptions indefinitely.

Incomplete learning from endogenous data is an established theme, including Brezzi and Lai's (2000) dynamic-allocation analysis. The comparison-budget contribution must be assessed as a particular object and theorem in that literature. This article adds an institutional account of how a decision-relevant experiment might be assembled when ordinary allocation does not produce it. The experiment can concern an input-loading parameter, a service choice, or a procurement decision, provided its target and observation channels are declared.

Independent replication is one possible response. Separate histories with a common structural parameter can generate discriminating evidence that a single reinforced history lacks. Whether independence is feasible depends on resource sharing, state retention, common shocks, and

response behavior. Merely resetting a displayed counter does not establish a new independent experiment. The institutional design must specify what is reset, what remains shared, and which uncertainty the new observations address.

Decision value then supplies the welfare bridge. Blackwell (1953) compares experiments by their usefulness in decision problems. An information gain has economic value through the actions it improves, not through an assumed price per comparison-budget unit. A high-stakes procurement decision and a trivial ranking adjustment can place very different values on the same discrimination improvement. The joint-permission model below therefore begins with a monetary decision value and derives a governance problem around its production.

4. Complementary evidence and a joint-permission game

Consider a one-time decision between actions zero and one. A state θ is equally likely to be zero or one. Choosing the wrong action costs $L > 0$. With no evidence, either fixed action has expected loss $L/2$. A calibrated binary signal can predict the state correctly with probability q , where $1/2 < q \leq 1$. Its expected loss is $L(1-q)$, so its gross decision value relative to no evidence is $V = L(q - 1/2)$.

Suppose the signal is available only through two complementary records held by different parties. Let X be a fair binary random variable independent of θ , and let E be an independent binary error with probability $1-q$ of being one. Define $Y = X \text{ XOR } \theta \text{ XOR } E$. Either X or Y alone is independent of θ ; jointly their exclusive-or predicts θ with accuracy q . This example establishes genuine informational complementarity: each record alone has zero decision value for this task, while the pair has value V . It does not assume that any real data pair has this structure.

Each holder i chooses whether to provide its record and the associated usable access, represented by $x_i=1$, or to withhold it, $x_i=0$. Providing incurs a real opportunity cost $c_i>0$ even if the other record is unavailable. Costs include all valued resource, security, and productive consequences included in the model. If both provide, holder i receives a share $\eta_i \geq 0$ of the gross decision value. Assume $\eta_1 + \eta_2 \leq 1$; the remainder benefits an outside decision beneficiary or an authorized sponsor acting for those beneficiaries. Baseline payoffs are normalized to zero.

$$u_i(x_1, x_2) = \eta_i V x_1 x_2 - c_i x_i. \quad (2)$$

Social incremental welfare counts the full decision value and all real provision costs. Payments among the parties will be transfers, not additional benefits. Before introducing a coordinator, welfare is therefore:

$$W(x_1, x_2) = V x_1 x_2 - c_1 x_1 - c_2 x_2. \quad (3)$$

Proposition 1. If $\eta_i V < c_i$ for each holder, withholding is strictly dominant and the unique Nash equilibrium is $(0,0)$. If also $V > c_1 + c_2$, joint provision is the uniquely efficient action profile.

The private result follows because provision gives $-c_i$ if the other holder withholds and $\eta_i V - c_i < 0$ if it provides. The social result follows by comparing the four action profiles: joint provision yields $V - c_1 - c_2 > 0$, withholding yields zero, and either unilateral provision yields a negative cost. Thus the equilibrium produces no evidence even though the joint experiment is socially worthwhile. The gap is an explicit appropriation failure combined with complementary control, rather than a conclusion inferred from concentration.

The assumptions allow holders to be entirely rational and productively successful. There is no need to assume hostility to a competitor or deceptive conduct. A holder can decline because it incurs a cost that its private share of the evidence benefit does not cover. In other cases the cost can be genuinely larger than the entire social benefit, making refusal efficient. The model

distinguishes those cases through V and the costs rather than through the identity or size of the holder.

If private returns exceed costs, joint provision may become a Nash equilibrium, but complementary provision can still leave a coordination problem because unilateral access remains costly. Bundling both permissions can remove separate veto points. It does not automatically eliminate incomplete appropriation: a bundled owner retaining only $\eta_1 + \eta_2$ of V can still find provision unprofitable. Property-right integration and funding are therefore analytically different remedies.

The result is deliberately elementary. Its contribution to this program is the explicit decomposition of an evidence-producing failure. A policy analyst can ask whether the inputs are actually complementary, whether the costs are real, whether the joint evidence is useful, and who benefits. Those questions are measurable or contestable. They are more informative than an unqualified assertion that successful firms crowd out every valuable counterfactual.

5. Committed contracting and the financing condition

Now introduce a coordinator that can offer an enforceable joint contract before either party supplies its record. The coordinator offers sequential, binding acceptance: the first holder accepts conditionally on the second accepting, and the second then decides. If either declines, the experiment is cancelled without provision costs. If both sign, audited provision becomes a contractual obligation; each holder receives a transfer T_i on satisfactory performance. The coordinator incurs an additional real implementation cost $H \geq 0$. Assume enforceability, verifiable provision, no distortionary cost of transfers, and an outside sponsor able to commit funds representing the residual decision beneficiaries. These are strong institutional assumptions, not conclusions of the model.

Focus on the underappropriation case $\eta_i V < c_i$. A holder participates weakly voluntarily when its transfer covers $c_i - \eta_i V$. At equality it is indifferent; strict participation requires a small additional payment. Minimum total compensation is $c_1 + c_2 - (\eta_1 + \eta_2)V$. The sponsor's gross benefit is $[1 - (\eta_1 + \eta_2)]V$. Its resources must cover the compensation and H .

$$\begin{aligned} T_1 + T_2 &\geq c_1 + c_2 - (\eta_1 + \eta_2)V, \\ [1 - \eta_1 - \eta_2]V &\geq T_1 + T_2 + H. \end{aligned} \quad (4)$$

Proposition 2. Under these contracting assumptions, a weakly voluntary, sponsor-financed joint contract exists if and only if $V \geq c_1 + c_2 + H$. If the inequality is strict, transfers can give each participant and the sponsor strictly positive net surplus.

Necessity follows by summing the holders' participation constraints and the sponsor's financing constraint. Sufficiency follows by choosing the holders' minimum transfers and funding them from the residual benefit. The remaining surplus is $V - c_1 - c_2 - H$. When positive, it can be divided to make participation strict. The feasibility result concerns committed obligations, not coordination by assumption. With strictly positive surplus allocated to both holders and the sponsor, the stated sequential binding-acceptance process induces both acceptances by backward induction. At the zero-surplus boundary only weak participation is established. Merely announcing success-contingent subsidies, or offering simultaneous all-or-nothing signatures, does not establish unique implementation. The appendix explains these differences.

This result locates the institutional burden. A successful coordinator must identify the joint experiment, value its decision consequences, verify provision, obtain commitment, and fund the gap. If the beneficiaries are dispersed and cannot pool resources, the model's sponsor may not exist. A public purchaser can sometimes perform this role for a procurement decision within its authority; a private buyer or consortium can perform it for its own task. Neither is presumed capable of representing every future beneficiary of market learning.

The permission costs may include losses in the holders' productive activity. Compensation does not make those costs disappear socially; they remain in the financing threshold. If access degrades service, discloses sensitive information, or consumes scarce frontier compute, a joint contract can fail the test. Likewise, if the signal cannot be calibrated or the beneficiaries would use it poorly, its V is lower than the nominal discrimination improvement might suggest. A defensible estimate should consider these possibilities rather than value information independently of its use.

Coase's (1960) analysis makes transaction costs and the institutional allocation of rights central to comparing arrangements. The present threshold gives an explicit transaction-cost term H for one evidence contract. It does not assume that private bargaining always solves the problem or that public coordination always lowers H . A public facility might reduce repeated negotiation costs while adding administrative delay, capture risk, or security expense. These enter the comparative institutional assessment as real effects.

When costs or value are uncertain, robust participation provides a conservative variant. Suppose holders' maximum credible costs are $\bar{c}_1$ and $\bar{c}_2$, the coordinator's maximum cost is $\bar{H}$, and a common evaluation rule guarantees decision value at least $\underline{V}$. A sufficient robust condition is $\underline{V} > \bar{c}_1 + \bar{c}_2 + \bar{H}$. This statement requires a common rule that works across the retained models; choosing a different ideal rule separately in each model overstates what an uninformed institution can implement. It is a sufficient condition, not an assertion that every experiment failing it has negative expected value.

6. Comparative institutional arrangements

Private bilateral or multilateral contracting is the first arrangement to examine. It can be efficient when a buyer captures much of the evidence benefit, costs are verifiable, and the required

permissions are easy to assemble. A contract can preserve ownership while leasing a bounded evaluation right. It can specify resource limits, access duration, privacy controls, and publication of aggregate findings. Its weakness appears when beneficiaries are diffuse, a holder's costs are difficult to verify, or future opportunities cannot contract in advance.

Public procurement offers a narrower coordination route. A purchaser chooses among systems for a specified public task and can finance evidence relevant to that choice. The purchaser need not establish the social value of every possible future entrant. It must define the task, eligibility standards, cost accounting, and the evaluation rule. A procurement experiment can then reveal whether a feasible alternative meets that task's requirements. Its result is a bounded performance claim, not a comprehensive estimate of a firm's market power or social contribution.

An independent testing facility can reduce repeated permission and setup costs across decisions. Shared compute, evaluation staff, secure records, and reproducible protocols can amortize fixed costs. The facility must still distinguish shared resources from statistically shared states.

Common infrastructure can support independent samples, but common hidden feedback can contaminate nominal replications. Independence should be established through the experimental design and recording rules rather than inferred from the facility's organizational label.

A testing commons raises its own governance questions. Ostrom (1990) shows why collective resource management requires attention to concrete rules and monitoring arrangements rather than a universal choice between market and state. The analogy concerns the design of a shared institution, not a direct identification of evaluation data with a common-pool resource. Secure records may be reusable, while compute, expert attention, and testing slots remain scarce.

Membership, allocation, contribution, appeal, and sanction rules must address those different resource properties.

Information design adds another caution. Institutions can change not only the amount of evidence but the signals presented to decision makers and the actions those signals induce.

Bergemann and Morris (2019) organize the economics of information design. For comparative discovery, a provider-controlled dashboard can reveal some distinctions while hiding others. An independent evaluator should specify the raw observation channel and the decision rule, including what is unavailable. A summary statistic is adequate only relative to the target for which it preserves the needed information.

No arrangement dominates without context. A private integrated owner can provide an excellent internal test when it captures the value and can credibly preserve independence. A public facility can be wasteful when the decision stakes are low or equivalent private evidence already exists. A plural supplier system can provide little new evidence if all systems are sampled through a common leader-biased gate. Institutional comparison should therefore rank feasible arrangements by decision value net of costs and incentive constraints, while retaining productive scale benefits.

An appeal procedure is also part of evidence production. Qualified alternatives need a way to challenge an eligibility error or a biased test, but an unrestricted right to demand expensive access would create its own inefficiency. A proportionate procedure can require a bounded prototype, a stated hypothesis, and a plausible decision affected by the evidence. It can then fund a limited trial or explain why the proposed comparison is uninformative or too costly. The institution protects a meaningful opportunity to test a claim, not an entitlement for every imagined firm to exist.

A worked example illustrates what comparative institutional assessment adds. All numbers are hypothetical monetary units for one bounded decision. Let $V=100$, $c_1=25$, $c_2=30$, $\eta_1=0.10$, and

$\eta_2=0.15$. Without compensation each holder prefers withholding, although joint provision has gross net social value 45 before coordination. The minimum transfers are 15 to each holder; the sponsor retains gross value 75. The same evidence and permission costs can therefore be worthwhile or not depending on the arrangement that assembles them.

Table 1. Hypothetical joint-permission arrangements for a decision with $V=100$

Arrangement	Coordination cost H	Net social value	Financing result
Ad hoc secure contract	50	-5	Not voluntary within the model
Procurement contract	30	15	Participation-feasible
Existing shared facility	20	25	Participation-feasible
New facility, first use	100	-55	Not worthwhile for this use

Permission costs total 55. Net social value is $100-55-H$; minimum holder transfers total 30. Differences in H are illustrative assumptions, not measured institutional advantages.

For the procurement contract the sponsor can pay the minimum compensation and $H=30$ while retaining 15. A small part of this surplus can make both holder acceptances strict. The existing facility yields more surplus only under its assumed lower total coordination cost and equivalent evidential quality. If provision cannot be verified, neither positive number establishes implementability; if a secure interface reduces signal accuracy, V must also be revised. The table therefore separates financing, enforceability, and informational adequacy.

The new-facility row includes a fixed setup cost of 80 plus per-use coordination cost 20. If there are m independent, equally valuable decisions, all costs and benefits are additive, and the alternative procurement arrangement costs 30 per use, facility welfare is $25m-80$ while repeated contracting yields $15m$. The facility is preferable only when $m>8$, with equality at eight uses. For one use it is harmful despite the attractive recurrent cost. Discounting, uncertainty about future use, and shared-state dependence can change this threshold. This example makes a concrete

institutional margin visible: reducing repeated coordination costs can warrant an infrastructure investment only when enough worthwhile and genuinely usable comparisons follow.

7. A bounded AI-service evaluation design

Consider a public organization deciding whether to adopt a model-assisted service for a defined class of tasks. The service boundary includes the task distribution, quality and reliability requirements, latency, total operating cost, human review, and security constraints. A feasible alternative is a demonstrated system that meets minimum technical and resource criteria, not a hypothetical company assigned unlimited compute. Productive capability qualification precedes the comparative trial.

The first stage documents the existing allocation mechanism. Record how tasks, users, or procurement opportunities reach providers; what state the mechanism retains; and what responses become observable. Administrative shares alone do not reveal conditional allocation probabilities. A comparison-budget estimate therefore requires an independently justified probability model or credible bounds. Any inference about the full observation experiment must include informative responses, cost measurements, and external evaluations already available.

The second stage specifies a feasible access counterfactual. It might be a secure test interface, a bounded compute allocation, a portability arrangement, or procurement opportunities allocated to an eligible alternative. The counterfactual should preserve the production requirements relevant to the task. If it changes the service or its resource envelope, that change becomes part of the intervention and its cost. The goal is to expose a particular disputed capability, not to guarantee adoption or permanent commercial survival.

The third stage produces independent or explicitly modeled task cohorts. Randomized exposure alone identifies no unknown parameter of a known assignment rule; the provider responses and

outcomes supply the relevant likelihood. The design should define which feedback is retained between tasks and cohorts, how common shocks enter, and whether learning within a model changes the response mechanism. A claim of independent replication requires common structural conditions with separately generated observations, or a model that handles their dependence. The fourth stage predefines the decision and evaluation rule. For procurement, the rule can combine error, reliability, human oversight, and cost under declared weights. A model comparison may instead target an input effect or an access-induced performance change. These are different estimands and can need different trials. The evaluator should report uncertainty regions, discrimination performance, and sensitivity to nuisance mechanisms rather than turn a single estimated comparison budget into a verdict.

The fifth stage values the decision improvement and full trial cost. A pilot has net value when better decisions outweigh compute, staff, delay, productive displacement, privacy, and implementation costs. The joint-permission model identifies which contract gaps might prevent a worthwhile pilot. A sponsor should also compare cheaper routes, including existing independent benchmarks, smaller targeted tasks, and secure sharing of already-produced evidence. A new facility is warranted only when its incremental benefit exceeds its incremental cost.

This design is prospective. No pilot has been run for this article and no empirical welfare number is supplied. A successful pilot would establish a bounded result under its task distribution and access conditions. Generalizing to an industry requires further evidence about demand, adaptation, investment, entry, and substitution. This limitation does not make the pilot trivial: public procurement is itself a consequential decision for which the bounded experiment can be appropriately designed.

8. From an epistemic diagnosis to policy instruments

An epistemic diagnosis begins with a target and an experiment. It asks whether the available allocation and response records can distinguish the alternatives relevant to that target. A comparison-budget theorem can provide a sufficient learning obstruction in its model; a finite-horizon likelihood or information analysis can diagnose more practical limits. Neither establishes that a legally relevant undertaking caused harm. The causal and welfare claims require additional evidence about an access intervention and its consequences.

The next step is an incentive diagnosis. Does the party controlling the comparison capture its net social value? The joint-permission model supplies one explicit negative answer. Other cases can involve a single controller, beneficiaries unable to contract, or complementary facilities with fixed costs. An institution that already internalizes all relevant benefits and costs has no underprovision problem in this benchmark. A remedy should be directed at the demonstrated incentive or contracting failure rather than at technological scale itself.

Instrument choice follows the diagnosis. A procurement contract can address a bounded purchaser's decision. An access obligation can change the evidence-generating mechanism, subject to authority and proportionality. A shared testing facility can reduce repeated setup costs. An experimental subsidy can address underappropriated benefits. A corrective activity charge needs a causal marginal effect on net external value and an administrable base. A general profit tax is not interchangeable with any of these instruments, and a tax cannot automatically supply a missing calibration or permission mechanism.

Pareto efficiency is also a stronger claim than a positive sum of net benefits. Proposition 2 gives a voluntary compensation arrangement within its closed model, but actual policies may leave uncompensated losers or face distortionary funding. A Pigouvian argument typically concerns

aligning private and social marginal incentives under stated conditions. It does not establish that a real company is undertaxed without measuring the relevant marginal effects and considering existing incentives. Professional policy analysis should report which efficiency criterion it uses and whose consequences it includes.

The approach can evaluate a specified access arrangement without taking a market-share power index as the primitive economic object. It cannot avoid all boundaries. A service, resource envelope, decision, eligible alternatives, and plausible substitutions must be stated. Existing antitrust law may separately require market definition, dominance, or other predicates. An economic framework can guide evidence gathering and the design of lawful procurement or evaluation instruments; it does not erase those requirements.

The policy contribution is consequently institutional and epistemic. It asks how rules for access and evaluation can preserve warranted decisions in technologies that generate scale advantages and adaptive evidence. It can recommend preserving productive integration while changing a test route. It can also recommend leaving an arrangement intact when the evidence gain is small, the costs high, or no feasible informative alternative exists. This capacity to reach opposing conclusions under different facts is essential to the framework's empirical credibility.

9. Evidence gates for a cumulative research program

The statistical gate is to verify the precise comparison-budget proposition and its observation scope. Formal verification can improve confidence in an encoded theorem, but conventional proofs remain valid scientific evidence and neither format verifies a market's premises. A rigorous audit should distinguish the main impossibility result from helper inequalities and additional information or predictive-merging claims. The empirical relevance of the result depends on the full observed channels and the adaptive mechanism.

The technological gate is to establish a feasible alternative. A proposed provider must satisfy a capability requirement under a bounded resource plan. The entry gate then asks how a specified access arrangement changes private viability. These tests distinguish expensive but efficient concentration from a modifiable restriction on qualified entry. They also permit the finding that no currently feasible alternative is suppressed, which would weaken the motivating policy hypothesis.

The measurement gate is to compare actual experiments rather than count nominal providers. Controlled designs, finite-horizon likelihoods, uncertainty calibration, nuisance sensitivity, and transparent computation are appropriate tools. Synthetic demonstrations can validate procedures against known truth, but actual market conclusions need actual records or trials. In particular, observed slow budget growth does not prove almost-sure finite total comparison; alternative reinforcement and state-retention mechanisms can produce different asymptotic behavior.

The institutional gate is to identify the permissions, resources, beneficiaries, and enforcement requirements for a worthwhile experiment. Proposition 2 supplies one financing threshold, conditional on a committed sponsor and verifiable costs. A field investigation should measure negotiation, privacy, security, productive, and administrative constraints, not assume them away. It should compare integrated, contracted, and shared arrangements before recommending a new public institution.

The welfare and instrument gate is to estimate decision value and private appropriation, then compare the least costly feasible interventions. Independent evidence can reveal that a trial has high discrimination value but little economic consequence. It can also reveal substantial consequence but no feasible low-cost remedy. A publication program should preserve those

negative findings. The objective is a cumulative, falsifiable account of the route from adaptive evidence to institutional choice, not a sequence of papers that assume the desired tax conclusion.

10. Conclusion

Scaling markets can concentrate production while leaving the efficient organization of comparative evaluation unsettled. The institutional question is which rules permit a feasible, informative experiment to be assembled and used. The joint-permission model shows a precise underprovision case and a committed financing condition: gross decision value must cover provision and coordination costs, while enforceable compensation aligns the holders' participation incentives.

The comparison-budget program supplies a reason to study these institutions when adaptive allocation changes the evidence available about alternatives. Moving from that diagnosis to policy requires technological feasibility, discriminating observation designs, measured decision value, and an incentive account. These obligations support an endogenous approach to contestability that can preserve productive scale, fund warranted comparisons, and decline costly interventions when their evidential benefits are insufficient.

Appendix A. Proofs and implementation boundaries

A.1. In the binary experiment, X is independent of θ by construction. For any fixed θ , the exclusive-or of the fair X with θ and E is also fair, so Y alone is independent of θ . Joint observation reveals $X \text{ XOR } Y = \theta \text{ XOR } E$. With an equal prior and $q > 1/2$, predicting θ by that exclusive-or is optimal and has error $1 - q$. The gross reduction in expected monetary loss is $L/2 - L(1 - q) = L(q - 1/2)$. Joint complementarity is thus demonstrated through a complete observation model rather than stipulated as an unspecified information benefit.

A.2. For holder i , provision changes its payoff by $-c_i$ when the other holder withholds and by $\eta_i V - c_i$ when it provides. Both are negative under Proposition 1, so withholding is strictly dominant. The four social payoffs are zero, $-c_1$, $-c_2$, and $V - c_1 - c_2$. Under $V > c_1 + c_2$, joint provision uniquely maximizes them. This proof requires real provision costs even for an unsuccessful unilateral release; different timing or recoverable costs produce a different game.

A.3. Under a committed joint contract, holder participation requires $\eta_i V + T_i - c_i \geq 0$. The sponsor's participation requires $(1 - \eta_1 - \eta_2)V - T_1 - T_2 - H \geq 0$. Summing gives $V - c_1 - c_2 - H \geq 0$, proving necessity. Setting $T_i = c_i - \eta_i V$ proves sufficiency in the maintained underappropriation case. Positive total surplus can be distributed among the three participants to make all inequalities strict. Enforcement and verifiable performance are assumptions of implementation; if a holder can collect payment without supplying a usable record, these inequalities do not secure the experiment.

A.4. A success-contingent subsidy offered in a simultaneous noncommitted game is different. When the other holder withholds, unilateral provision still yields $-c_i$, so $(0,0)$ remains an equilibrium even if subsidies make $(1,1)$ attractive. Simultaneous all-or-nothing signing can likewise retain a no-signing equilibrium: if the other holder declines, signing makes no difference. Proposition 2 establishes participation-feasible contract existence. Under strictly positive distributed surplus, sequential binding acceptance adds an implementation result by backward induction: the second holder accepts after the first, and the first anticipates that acceptance. Neither enforceability alone nor a subsidy announcement proves unique implementation. If the coordinator cannot form or enforce that contract, it must model the remaining coordination failure and its cost.

A.5. Full institutional comparison also requires financing incidence. Transfers cancel in the stated social objective because holders and sponsor belong to the same welfare population and funding is nondistortionary. If public revenue has a shadow cost, some beneficiaries are outside the population, or transfers change incentives for other activities, those consequences must be added. Likewise, a stochastic evidence value requires a common decision rule and a specified expectation or robust criterion. The scalar V is valid for the fully defined binary example; its empirical use requires those additional choices.

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Declarations

This article is a theoretical institutional analysis with a prospective evaluation design. It contains no proprietary empirical data and reports no completed field trial. Its illustrative game and signal construction are fully specified in the text. Substantive AI assistance through OpenAI Codex was used to develop and revise prose, mathematical arguments, reference checks, and document production. The author is responsible for the manuscript.